

SIDDHARTH PRABHU

AI Engineer · Edge & Production ML

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SUMMARY

AI Engineer focused on production-grade systems that run reliably outside the notebook — RAG pipelines, edge computer vision, and agentic workflows. At nhance.ai, cut enterprise document turnaround by **90%**, served RAG over **10k+** documents at **sub-250ms p95**, and ran edge vision at **30 FPS** with **40% lower latency**. IEEE-published, edge-first, and obsessive about latency, reliability, and systems that don't hallucinate.

TECHNICAL SKILLS

Languages: Python, JavaScript, SQL, Bash

GenAI & LLM: LangChain, RAG, Agents (ReAct, multi-agent), Prompt Engineering, Gemini, OpenAI, Llama 3 (Groq / local), FAISS, Milvus

Vision & Edge: YOLOv8 / v11, OpenVINO, OpenCV, BoT-SORT, PyTorch, TensorFlow, CNN / LSTM / RNN

Backend & Systems: FastAPI, Node.js, React, REST APIs, MongoDB, asyncio, WebSockets, microservices

DevOps & Tools: Docker, Git, CI/CD, Linux (Fedora / Ubuntu), n8n

EXPERIENCE

nhance.ai

Sept 2024 – Present

Data Science & AI Engineer

Bengaluru, India

- Architected an end-to-end **MERN** configurator automating SOW, BOQ, BRD, and proposal generation, cutting document turnaround from hours to minutes (**90%+** reduction).
- Built an edge video-analytics pipeline ingesting **5 concurrent RTSP streams at 30 FPS**; cut inference latency **150ms → 90ms (40%)** through OpenVINO optimization.
- Shipped a production RAG assistant over **10k+** technical documents using Milvus hybrid retrieval, holding **p95 latency under 250ms**.
- Designed multimodal document-intelligence workflows with Gemini for summarization, Q&A, and structured extraction across PDFs and images.

M.Tech Solutions

Aug – Sept 2024

Information Security Intern

Bengaluru, India

- Automated threat analysis across **40+** security datasets in Python, improving assessment turnaround by **25%**.
- Hardened access-control infrastructure through Role-Based Access Control (RBAC) implementation.

SELECTED PROJECTS

Real-Time Voice Agent (Jarvis) | *asyncio, Deepgram, Groq Llama 3.1*

[repo](#)

- Full-duplex voice assistant at **<500ms latency** on an asyncio triple-loop architecture (listen / think / speak) coordinated by lock-free thread-safe queues, streaming Deepgram STT + Aura TTS around a Groq Llama 3.1 token stream.
- Engineered a smart TTS buffer (punctuation / 30-char flush) for **<300ms** perceived latency, plus an RMS noise-gate and minimum-length speech filter to reject false barge-ins from background noise.

Edge Video-Analytics Pipeline | *YOLOv11, BoT-SORT, OpenCV*

[repo](#)

- Self-healing people-counting system built as distributed workers under a central supervisor that auto-restarts any crashed or frozen process; precise line-crossing gate logic for entry/exit counts.
- Exposes a live JSON metrics API and a responsive dashboard with real-time CPU / GPU / memory health monitoring; configurable via YAML with env-var secret handling.

Multimodal Document QA | *FastAPI, Milvus, Tesseract, Gemini*

[repo](#)

- Document QA service with a custom ingestion pipeline that auto-detects file types — PDF text extraction, Excel row parsing, OCR on images — exposed via REST `/upload` and `/query` endpoints and containerized with Docker.

PUBLICATION & RECOGNITION

IEEE ESCI 2024 — Published *Image Processing Based Traffic Surveillance*; trained a custom CNN on **5,000+** annotated frames achieving a **92% F1-score**. · **NCC Sergeant** — led team coordination and discipline-focused programs. ·

Writer — @algorithm-chemist on Medium (RAG, vector DBs, MLOps).

EDUCATION

Siddaganga Institute of Technology

2020 – 2024

B.E. in Electronics & Communication Engineering

Tumakuru, Karnataka